Bank Branch Network Contraction and Employer Startup Activity in North Carolina Counties, 2014-2023

Aadi Joshi

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Abstract

This paper examines the association between full-service bank-branch density and employer-startup activity across North Carolina counties. I assembled a reproducible panel from the Federal Deposit Insurance Corporation Summary of Deposits, U.S. Census Bureau Business Dynamics Statistics (BDS), Bureau of Economic Analysis, Bureau of Labor Statistics, and USDA. The candidate panel covers every North Carolina county from 2014 through 2023. Five disclosure-suppressed startup cells reduced the main estimation sample from 1,000 to 995 county-years. County and year fixed-effects models adjust for unemployment, population, and per-capita income. North Carolina's full-service branch count fell 21.8 percent between 2013 and 2022. In the internally frozen specification, a June 30 branch snapshot labeled year t-1 is matched to a BDS flow labeled year t. Because the BDS flow runs from March t-1 to March t, the snapshot falls inside the outcome interval. The estimated coefficient is −1.823 startup establishments per 1,000 lagged establishments for one additional branch per 10,000 residents (95% CI −5.412 to 1.767). A specification added during the audit moves the exposure and controls back another year, placing them before the outcome interval; its estimate is 2.028 (95% CI −1.383 to 5.439). Both estimates are imprecise, their signs differ, and neither has a causal interpretation. The panel does not establish a precise association with a stable direction across timing choices. It contributes reproducible state-level measurement and documents the limits of what these data can show.

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Research status and assistance disclosure. I wrote this manuscript. Generative AI was used to review and revise selected passages and to assist with technical and quality checks. I verified the analysis and take responsibility for the final text and conclusions.

1 Introduction

The number of **physical** bank branches in the United States has fallen substantially since 2010. That trend led me to ask a question about communities like mine: when the banking network contracts, does the formation of new employer businesses change? Young firms often seek outside financing before they have long financial histories, established collateral, or public credit ratings. A nearby banker can gather qualitative information that does not fit neatly into a standardized application, which gives physical offices a possible role in relationship lending. Online applications, bank information technology, fintech platforms, credit scoring, and travel across county lines complicate that story. A branch count alone cannot capture access to finance.

North Carolina provides substantial variation within one state. It contains fast-growing metropolitan counties, small towns, and rural counties with very different population densities and banking markets. In the assembled data, qualifying full-service offices fell from 2,554 in 2013 to 1,998 in 2022, a decline of 21.8 percent. Statewide branch density fell from 2.606 to 1.865 branches per 10,000 residents, a steeper decline because population increased while the branch count contracted. Employer startup activity did not move mechanically with the branch network. The constructed statewide startup rate was 57.0 per 1,000 lagged establishments in 2014 and 71.6 in 2023, with a pronounced pandemic-period pattern. These series describe the state; they do not separate branch conditions from the economic shocks that affected every county during the decade.

This paper asks:

> How is full-service bank-branch density associated with employer-startup activity in North Carolina counties from 2014 through 2023, after accounting for time-invariant county characteristics and common year shocks?

The central word is **associated**. Managers may close a branch because they expect weaker local growth or lower future loan demand. Employment, industry composition, migration, housing markets, and disasters can influence both branch decisions and business formation. County and year fixed effects, together with three time-varying controls, improve on a simple cross-sectional comparison. They do not create an experiment. The headline model has an additional timing problem: the June 30 SOD snapshot falls inside the March-to-March BDS transition labeled t. The analysis therefore does not estimate the causal effect of closing a branch.

I build a public, reproducible North Carolina panel from five federal data systems and measure employer startups with BDS firm-age-zero establishments. That choice separates establishments created by new employer firms from new locations opened by firms that already exist. The paper records the model sequence fixed before I inspected the estimates, how I handled disclosure suppression, and the results of subsequent checks. Its policy value lies mostly in clarifying what the data can establish. Branch density is not geographic access, and an imprecise county-level estimate

cannot show that no neighborhood or borrower is harmed by a closure.

The headline estimate is negative and imprecise. In the internally frozen model, one additional full-service branch per 10,000 residents is associated with 1.823 fewer startup establishments per 1,000 lagged active establishments (county-clustered standard error 1.809; 95 percent confidence interval -5.412 to 1.767; $N = 995$). I retain that model as the original primary specification because I chose it before inspecting the estimates. A model added during the audit moves the exposure and controls before the outcome interval and instead estimates 2.028 (standard error 1.719; 95 percent confidence interval -1.383 to 5.439; $N = 895$). Neither interval excludes zero. The estimates do not point consistently in either direction. Metro and nonmetro slopes, alternative outcomes, and most robustness checks are likewise imprecise. One Poisson count specification produces a negative conditional association with an interval excluding zero; I retain it as a model-dependent secondary result rather than treating it as decisive.

The remaining sections describe the institutional setting, data construction, empirical strategy, results, robustness checks, and policy limits. The appendices record the sample flow, flag treatment, equations, quality checks, and supplementary estimates.

2 Institutional Background and Literature

2.1 Relationship lending, distance, and entrepreneurship

Small-business finance involves an information problem. A young firm may lack audited statements, a long repayment history, or assets suited to standardized underwriting. Through repeated contact, a lender can learn about the owner, local demand, suppliers, and the firm's ability to execute. Physical proximity may lower the cost of collecting this "soft" information. Technology can make some of this information portable.

Petersen and Rajan (2002) document that the distance between small firms and their lenders increased as communication became less personal and lender productivity improved. The finding leaves room for physical proximity to matter while showing that its role in credit provision changes with technology. Interpreting recent branch counts requires allowing for borrowers who do not meet a banker in person.

Historical banking deregulation offers broader evidence that finance can affect entry. Using Census microdata and differences in state deregulation timing, Kerr and Nanda (2009) find increases in both entrepreneurship and business closures after banking reforms. They interpret this combination as evidence that broader access allows more experimentation, including ventures that later fail. Their treatment, period, and design differ from the recent contraction of branch networks, but the study establishes a plausible link between credit-market structure and new-firm formation.

The distinction between new firms and new locations matters here. Haltiwanger, Jarmin, and Miranda (2013) show that firm age, rather than small size alone, is central to job-creation patterns. Decker et al. (2014) document both the importance of startups and young firms to U.S. economic dynamism and the long-run decline in startup activity. These findings do not imply that every

startup becomes a high-growth firm or that more entry is always better. They do support measuring establishments created by genuinely new employer firms rather than counting every new location.

2.2 Branch closures and real activity

The most persuasive prior work uses data and identification strategies unavailable in public county aggregates. Nguyen (2019) studies branch closings during the 2000s and instruments for them with tract-level exposure to post-merger consolidation. The paper finds persistent declines in nearby small-business lending, concentrated within a short distance. That design addresses why banks choose particular branches to close more directly than the county fixed-effects model used here, and it measures geography more precisely.

Recent Federal Reserve research extends the question to business outcomes. Ranish, Stella, and Zhang (2024) combine merger-related closure variation with restricted Longitudinal Business Database records and report reductions in small-business employment growth and entry near closures. This is the closest modern benchmark for the question posed here, but the public panel cannot replicate it. It uses a smaller geographic unit, measures nearby closures directly, and draws on microdata and merger-related variation unavailable in this panel. That paper, along with the Ahnert et al. and Adams-Sherlund studies cited below, belongs to the Federal Reserve's Finance and Economics Discussion Series. These are preliminary staff working papers that express their authors' views and do not indicate concurrence by the Board of Governors.

Changes in lending do not always translate into clear changes in business activity. Greenstone, Mas, and Nguyen (2020) use predicted county lending shocks and find that negative shocks reduce loan originations. Their estimated effects of small-business lending on small-firm and overall employment, however, are generally small and statistically insignificant. Even with a stronger design, a measurable change in credit supply may have little detectable effect on aggregate employment. Substitution across lenders, internal finance, other credit channels, firm composition, and measurement can all weaken the link.

2.3 Digital banking and the meaning of access

Physical networks form only one part of financial intermediation. Ahnert et al. (2024) find stronger young-firm job creation in counties more exposed to banks with greater information-technology adoption. Their evidence is consistent with better collateral evaluation and faster loan processing. Digital capability may replace some branch functions, complement relationship bankers, or benefit particular borrowers. A lower branch count therefore need not produce an equally large decline in effective credit access.

Branch counts and distance can also move differently. Adams and Sherlund (2025) construct granular block-level distance measures for 2014-2024. They report that a 19 percent decline in the national branch count was accompanied by only modest increases in average distance to the nearest branch, despite meaningful losses in some places. A county can lose geographically redundant offices, or retain the same number while residents move farther from them. I therefore use **branch density** to describe the exposure; it does not measure access or identify banking deserts.

Rural settings may still experience branch loss differently. Federal Reserve listening sessions and descriptive analysis document acute difficulties in some rural communities that had few alternatives and long travel distances (Board of Governors of the Federal Reserve System 2019). That evidence motivates a metro/nonmetro interaction but does not justify predicting a uniform rural effect. Rural counties differ in size, adjacency to cities, broadband, transportation, credit unions, and local industry. A binary county classification cannot capture all of these differences.

2.4 Contribution and expectations

Existing research uses restricted national microdata and quasi-experimental designs to study local branch closures and small-business outcomes. This paper takes a narrower route: a public, reproducible county-level application for North Carolina. Census firm-age-zero tabulations distinguish establishments created by new employer firms from all establishment entrants, and a prespecified interaction examines metro and nonmetro differences. The public data make the measurement reproducible, but they cannot support the causal claims available from those stronger designs.

Following the relationship-lending mechanism, I expected higher branch density in the snapshot labeled $t-1$ to be positively associated with startup activity in BDS label t, perhaps more strongly in nonmetro counties. I timestamped the plan before inspecting the estimates. I did not register it externally and do not describe it as preregistered. During the later audit, I found that matching these year labels places the branch snapshot inside the BDS flow rather than before it. Digital substitution, cross-county borrowing, endogenous changes in bank networks, county aggregation, and the timing overlap all leave room for a small or unstable estimate. A precise negative coefficient would run counter to the directional hypothesis. An imprecise interval containing zero cannot confirm or reject the mechanism.

3 Data and Measurement

3.1 Sample and source alignment

The unit of analysis is a North Carolina county in a BDS outcome year. Outcomes cover 2014-2023. Banking exposures and controls cover 2013-2022 and are matched by label to BDS outcome years 2014-2023. The candidate panel contains all 100 counties in each of ten years, or 1,000 county-years. Five disclosure-suppressed age-zero establishment cells reduce the primary model to 995 observations.

The BDS county file supplies active establishments, firms, employment, and official entry and exit rates. A second BDS file, organized by county and coarse firm age, supplies establishments, firms, and jobs created by births for firm age zero (U.S. Census Bureau 2025, 2026). The raw 2023 time-series release contains 146,924 county rows and 734,620 county-by-age rows nationally. I remove eleven North Carolina residual-code rows with county code `999` in 2013-2023, leaving 1,100 true county denominator rows. Extracting the firm-age-zero records yields exactly 1,000 candidate county-years.

The FDIC Summary of Deposits (SOD) is an annual survey of branch-office deposits as of June

30 (Federal Deposit Insurance Corporation 2026). Annual North Carolina API extracts for 2012-2022 contain 26,330 branch records. Data from 2012 enter only the one-year change measure used in the 2014 model. BEA CAINC1 provides county population and nominal per-capita personal income. BLS Local Area Unemployment Statistics supplies annual-average county unemployment, and USDA Economic Research Service Rural-Urban Continuum Codes provide the 2013 and 2023 metro classifications (U.S. Bureau of Economic Analysis 2026; U.S. Bureau of Labor Statistics 2026; USDA Economic Research Service 2025).

The frozen model matches the June 30 SOD snapshot labeled $t-1$ to the BDS transition labeled t. Census defines BDS employment from the payroll period that includes March 12 and defines annual flows as changes from March $t-1$ to March t (U.S. Census Bureau 2026). The June 30 snapshot therefore occurs inside the outcome interval. This is why I retain the original model as a label-aligned comparison rather than describe it as predicting a subsequent outcome. An audit-added specification uses the SOD snapshot and controls labeled $t-2$, which precede the March $t-1$ start of the BDS interval.

3.2 Primary outcome: employer-startup rate

Census assigns an establishment the age of its parent firm. A firm-age-zero cell contains establishments entering as part of a new employer firm (U.S. Census Bureau 2026). The primary outcome is

$$StartupRate_{it} = 1{,}000 \times \frac{AgeZeroEstablishments_{it}}{AllActiveEstablishments_{i,t-1}}.$$

The resulting units are employer-startup establishments per 1,000 lagged active establishments. Using the lagged denominator scales the flow by the preexisting local business base and prevents same-year entrants from enlarging their own denominator. This rate is constructed for the study; it is not an official BDS variable.

The measure is narrower than "entrepreneurship." BDS covers employer businesses and excludes nonemployers. It does not count applications, incorporations without payroll employment, or individual founders, nor does it identify venture-backed or high-growth firms. Its main advantage is conceptual: it distinguishes establishments created by new employer firms from expansions by existing firms.

Five `age0_estabs` cells carry a `D` disclosure-suppression flag: Camden County in 2019 and Tyrrell County in 2015, 2017, 2021, and 2023. I preserve them as missing rather than recoding them as zero. The cleaning code applies the same principle to `S` (data-quality suppression), `X` (structurally zero or missing), and `N` (a rate unavailable because its denominator is zero). Only `D` and `N` occur in the retained years and variables, and all five missing primary-outcome cells are flagged `D`.

Two secondary outcomes broaden the comparison. The official BDS establishment-entry rate includes locations opened by existing firms. Startup job-creation intensity equals 1,000 times jobs created by firm-age-zero establishment births divided by lagged county employment. I do not use

the age-zero subset's own entry-rate field because its denominator makes that rate mechanically uninformative for this purpose.

3.3 Branch measures and controls

The main exposure retains SOD service codes 11 (full-service brick-and-mortar) and 12 (full-service retail). It excludes cyber, mobile, home-banking, and limited-service offices. For county i and exposure year $t-1$,

$$BranchDensity_{i,t-1} = 10{,}000 \times \frac{FullServiceBranches_{i,t-1}}{Population_{i,t-1}}.$$

In the assembled panel, branch density ranges from 0.489 to 8.373 branches per 10,000 residents and has a mean of 2.449. Every county has at least one qualifying branch in every exposure year. Branch counts vary over time in 92 counties. Comparing the endpoints, 88 counties have fewer branches in 2022 than in 2013, 11 are unchanged, and one has more.

Branch density is a ratio rather than a location-based access measure. The numerator omits credit unions, nonbank community-development financial institutions, fintech lenders, and other nonbank providers; the denominator moves with population. The variable says nothing directly about distance, drive time, hours, services, service quality, relationship continuity, or options across county lines.

The three predefined controls are annual unemployment, log population, and log nominal per-capita personal income under the same label convention as the exposure. This small set captures broad local economic change without adding a long list of variables that might themselves respond to changes in banking conditions. Year fixed effects absorb statewide price-level changes shared across counties; nominal PCPI is not interpreted by itself as a real-income series over time.

Deposit concentration is a secondary banking measure. For institutions b operating in county i, the HHI is $10{,}000 \sum_b s_{bit}^2$, where s is the institution's share of deposits assigned to county branch offices. Deposits booked at a branch need not belong to residents or businesses in the same county. I estimate HHI separately from branch density. The plan listed historical community-bank share as optional, but the data I acquired did not include a consistent institution-year classification. I omit the measure instead of constructing a new classification after observing the results.

The primary rural indicator assigns USDA 2013 RUCC codes 1-3 to metro and 4-9 to nonmetro. Fifty-four counties are nonmetro under that vintage. The 2023 version classifies 55 counties as nonmetro, and seven counties change status between vintages. I use the 2013 classification because it predates the outcome window and treat 2023 status as a robustness check.

3.4 Descriptive statistics and sample flow

Across the 995 observed county-years, the primary outcome has a mean of 57.672, a standard deviation of 18.116, and a median of 56.738. The mean age-zero establishment count is 139.8, although the distribution is strongly right-skewed because metropolitan counties are much larger.

The remaining summaries use all available observations for each variable. The official BDS entry rate averages 8.983. Startup job-creation intensity averages 20.271 per 1,000 lagged workers. Mean unemployment is 5.772 percent, mean county population is 102,253, and mean nominal PCPI is $41,148 under the frozen label alignment.

The sample-flow audit begins with 1,000 candidate county-years and loses five to BDS disclosure suppression. The core banking, control, and RUCC merges drop no additional observations. The repository records raw-file hashes, row counts, flags, and merge indicators. Figure 1 maps changes in branch density; Figure 2 presents statewide trends. Both figures are descriptive, not estimates of causal effects.

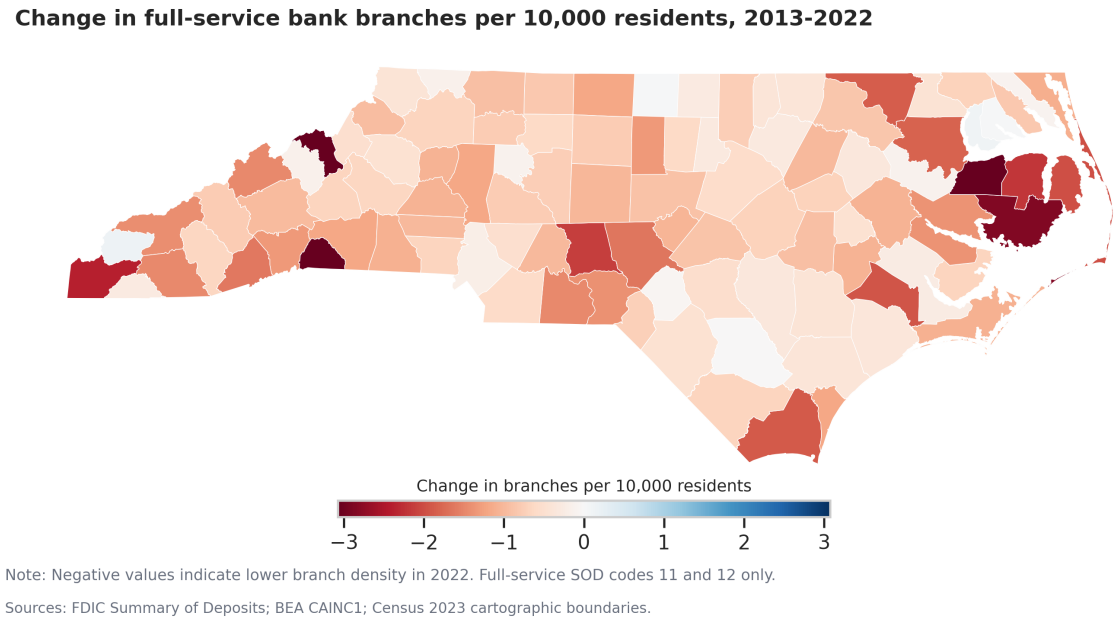


Figure 1: County changes in full-service branch density

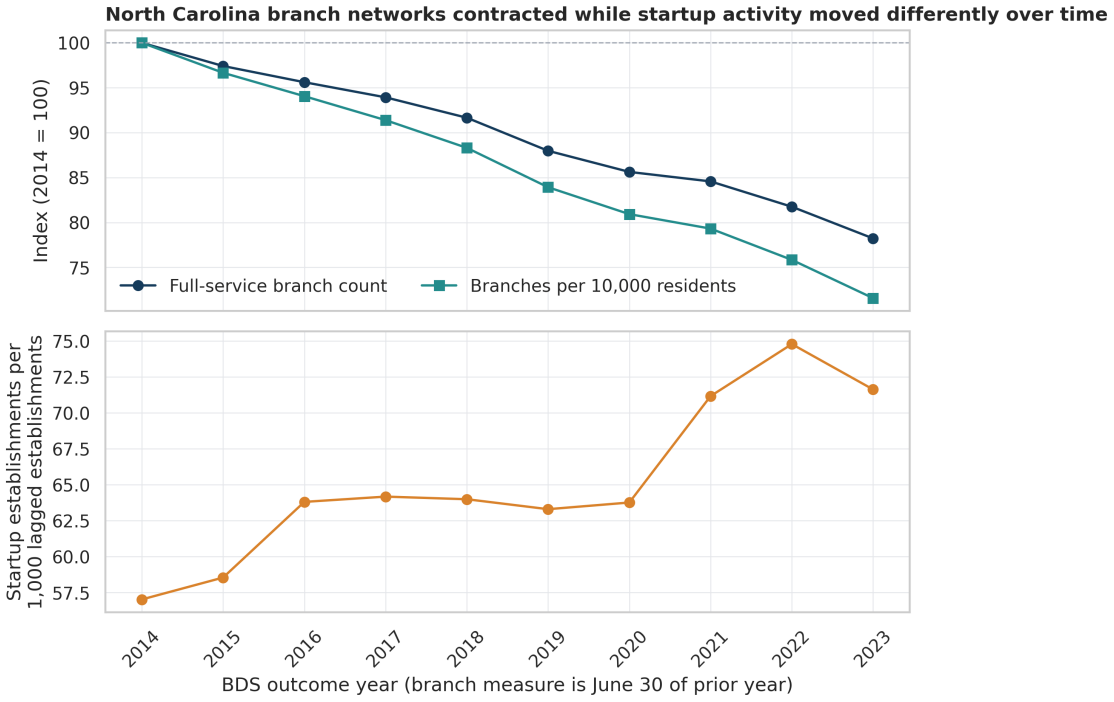


Figure 2: Statewide branch and startup trends

4 Empirical Strategy

4.1 Fixed-effects model

The headline equation is

$$Y_{it} = \alpha_i + \lambda_t + \beta B_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{it},$$

Here, Y_{it} is the employer-startup rate; $B_{i,t-1}$ is full-service branch density under the dataset-label alignment; $X_{i,t-1}$ contains unemployment, log population, and log PCPI under that same alignment; α_i is a county fixed effect; and λ_t is an outcome-year fixed effect. The subscript records the file-year match; as explained above, the June snapshot falls within the March-to-March outcome interval.

County fixed effects absorb time-invariant differences such as persistent geography, baseline urbanity, and stable industrial characteristics. Year fixed effects absorb shocks shared by North Carolina counties in a given year. The coefficient comes from changes in branch density within each county, after accounting for common year shocks and the three covariates. Persistent differences between counties with many branches and counties with few do not determine the estimate.

I cluster standard errors by county, allowing heteroskedasticity and serial correlation within each of the 100 county clusters. I report the coefficient in its original units as the primary estimate. For scale, I also multiply the slope by the within-county standard deviation of branch density in the estimation sample.

4.2 Rural heterogeneity

The heterogeneity equation is

$$Y_{it} = \alpha_i + \lambda_t + \beta B_{i,t-1} + \theta(B_{i,t-1} \times Nonmetro_i) + \gamma' X_{i,t-1} + \varepsilon_{it}.$$

County fixed effects absorb the time-invariant nonmetro main effect. The metro slope is β, the nonmetro slope is $\beta + \theta$, and θ is the difference between them. I report all three quantities with confidence intervals. An imprecise interaction cannot show that rural counties are unaffected.

4.3 Internally frozen model sequence and interpretation rule

I wrote and timestamped the analysis plan internally before inspecting the regression estimates. It was not externally registered. The sequence begins with county and year fixed effects without controls, then adds the three controls, the rural interaction, two secondary outcomes, first differences, exclusion of 2020 and 2021, population weighting, alternative branch exposures, deposit HHI, exit context, large-county influence checks, and a Poisson count model if it can be estimated reliably. I did not add machine learning or a merger instrument after seeing the results. One planned secondary outcome, firm-age-zero firms per 1,000 lagged firms, was inadvertently omitted from the original run. I estimated it during the audit as A1 and label it audit-added rather than treat it as a result from the original sequence.

Every estimate is a conditional association. Using an earlier exposure year does not eliminate reverse causality: a county-specific demand shock can reduce both branch presence and startups. Population growth can also lower branches per capita even when the branch count changes little. Measurement error and aggregation within counties may attenuate or distort the slope. This fixed-effects model is a transparent descriptive design, not a substitute for the merger-based or microgeographic designs used in prior work.

5 Results

5.1 Branch contraction and the headline estimate

The branch network contracted across most of the state. Qualifying offices fell by 556 between 2013 and 2022, and 88 counties ended the period with fewer branches. Statewide density declined 28.4 percent. The map also shows considerable variation in per-capita change; in a small county, one office can move the density measure sharply.

Table 1 reports the main estimates. With county and year fixed effects but no time-varying controls, the branch-density slope is -1.418 (95 percent CI -5.121 to 2.285). Adding unemployment, log population, and log PCPI under the frozen label alignment moves the slope to -1.823 (95 percent CI -5.412 to 1.767; $p = 0.316$). The controls neither reverse the sign nor materially improve precision.

Specification / outcome	Reported slope	Estimate	Clustered SE	95% CI	N
FE only: startup rate	Full sample	−1.418	1.866	[−5.121, 2.285]	995
FE + controls: startup rate	Full sample	**−1.823**	1.809	**[−5.412, 1.767]**	995
Rural interaction: startup rate	Metro	1.575	2.921	[−4.220, 7.370]	995
Rural interaction: startup rate	Nonmetro	−2.312	1.818	[−5.919, 1.295]	995
Official BDS entry rate	Full sample	−0.210	0.222	[−0.650, 0.230]	997
Startup job-creation intensity	Full sample	−1.223	1.387	[−3.976, 1.529]	995

Table 1. Selected fixed-effects estimates. Notes: Exposure is one additional full-service branch per 10,000 residents. The frozen models align an SOD snapshot labeled $t-1$ with a BDS flow labeled t; that snapshot partly overlaps the March-to-March outcome interval. Every model includes county and outcome-year fixed effects; all except the first include unemployment, log population, and log PCPI under the same label alignment. Confidence intervals use county-clustered standard errors. Source: author calculations from FDIC SOD, Census BDS, BEA, BLS, and USDA data. Full code-generated table: `output/tables/table_primary_estimates.csv`.

The point estimate runs opposite to my initial prediction, but the confidence interval includes moderately negative and moderately positive associations as well as zero. The result is **inconclusive and does not support the directional hypothesis**. It establishes neither a negative relationship nor the absence of a relationship.

The within-county standard deviation of branch density is 0.409. Multiplying that value by the headline coefficient gives −0.745 startup establishments per 1,000 lagged establishments, or −1.29 percent of the sample mean. This calculation puts the estimate in perspective without changing its uncertainty or resolving why branch density changed.

5.2 Metro and nonmetro estimates

The interaction using 2013 RUCC status does not establish a metro/nonmetro difference. The estimated metro slope is 1.575 (95 percent CI −4.220 to 7.370); the nonmetro slope is −2.312 (95 percent CI −5.919 to 1.295). Their difference, the nonmetro-minus-metro interaction, is −3.888 (95 percent CI −9.534 to 1.759; $p = 0.175$). The point estimates differ, but the uncertainty is too large to conclude that the association is negative in nonmetro counties or more favorable in metro counties.

Using 2023 RUCC status does not resolve the uncertainty. The metro slope is 2.368 (95 percent CI −1.499 to 6.236), and the nonmetro slope is −2.462 (95 percent CI −6.226 to 1.302). Changing the classification of the seven counties whose status differs between vintages does not establish a difference in the slopes.

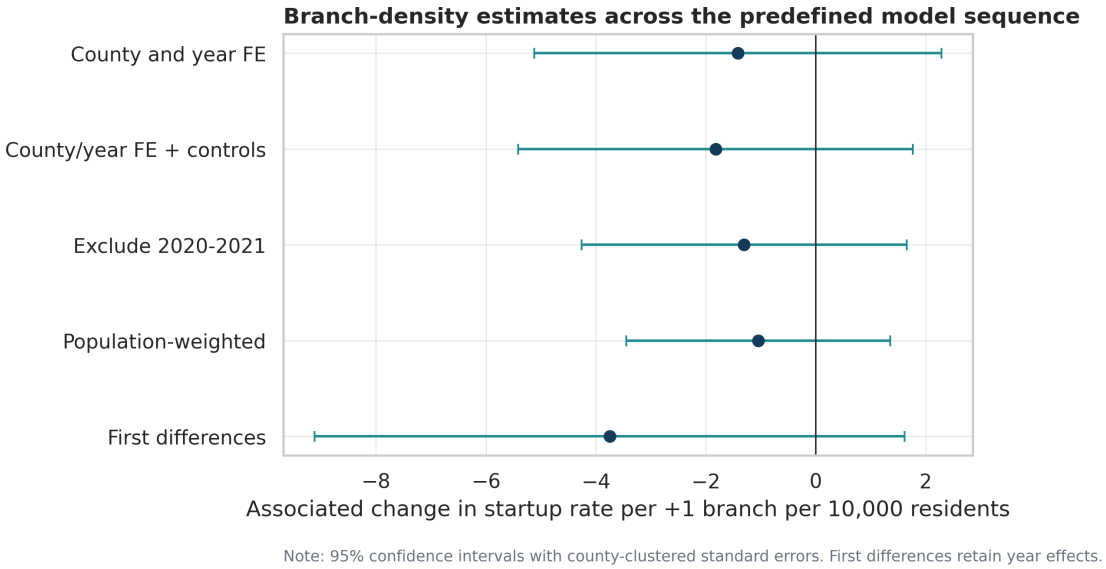


Figure 3: Predefined branch-density estimates

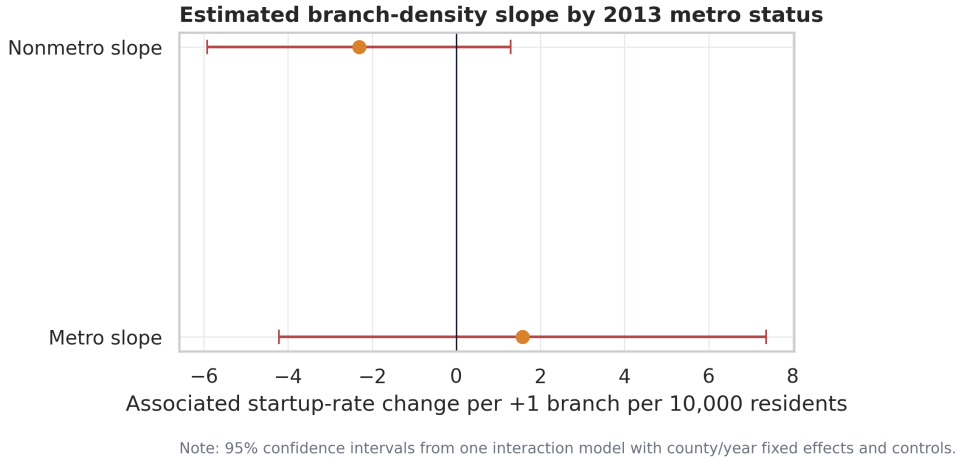


Figure 4: Metro and nonmetro branch-density slopes

5.3 Alternative outcomes

With the official BDS establishment-entry rate as the outcome, the slope is −0.210 (95 percent CI −0.650 to 0.230; $N = 997$). This official measure includes expansions by existing firms and therefore captures a broader form of establishment dynamism. Using startup job-creation intensity instead gives −1.223 jobs created per 1,000 lagged workers (95 percent CI −3.976 to 1.529; $N = 995$). Neither estimate is positive or precise.

The negative point estimate appears across more than one outcome. These are related measures from the same county-level BDS system, however, and all use the same branch exposure. Their similar signs do not provide independent confirmation.

6 Robustness and Heterogeneity

The predefined robustness checks do not resolve the uncertainty in the main model. Excluding outcome years 2020 and 2021 gives −1.302 (95 percent CI −4.260 to 1.656; $N = 796$). Population weighting gives −1.048 (95 percent CI −3.449 to 1.352). First differences produce −3.749 (95 percent CI −9.113 to 1.615; $N = 891$). Every interval includes zero.

The estimates also depend on how branch presence is measured. Replacing per-capita density with lagged branch count yields −0.001 (95 percent CI −0.313 to 0.310), essentially zero in its natural units. The one-year change in branch count gives −0.130 (95 percent CI −0.559 to 0.300). The one-year density-change estimate is more negative at −4.216, although its interval still includes zero (95 percent CI −8.948 to 0.517; $p = 0.080$). These checks were predefined. I do not select among them according to the smallest p-value.

Within counties, branch density correlates −0.682 with log PCPI, 0.407 with unemployment, and −0.123 with log population. These variables move together, though they are not perfectly collinear. The FE-only slope (−1.418) and controlled slope (−1.823) nonetheless lead to the same substantive conclusion. The separate HHI model examines deposit concentration without adding another banking measure to the headline equation.

For deposit HHI, scaled so that the coefficient represents 1,000 HHI points, the estimated slope is 0.636 (95 percent CI −1.503 to 2.775). Using the official establishment-exit rate as the outcome gives 0.280 (95 percent CI −0.183 to 0.743). Neither result is precise. When I omit Wake, Mecklenburg, Guilford, Forsyth, or Cumberland County one at a time, the headline branch-density coefficient remains between −1.821 and −1.790. No single large county accounts for it.

The Poisson model uses the age-zero establishment count, includes log lagged establishments as an offset, and retains the same fixed effects and controls. Its coefficient on branch density is −0.0404 (95 percent CI −0.0765 to −0.0043; $p = 0.028$), equivalent to an incidence-rate ratio of 0.960 per additional branch per 10,000 residents. This is the only predefined specification with a conventional 95 percent interval that excludes zero. Omitting it would be selective, so I report it. I retain it as a secondary result because it models expected startup counts differently from the rate regression while using the same observational exposure. Its negative sign also runs opposite to the theoretical

prediction. An interval excluding zero in one of many correlated specifications does not establish a causal result.

6.1 Post-hoc diagnostics

After completing the frozen sequence, I asked whether the population denominator could explain why the branch-count estimate was near zero while the density estimate was negative. Because branch density divides by population, population growth could contribute mechanically to the negative slope. I specified four diagnostics after seeing the original estimates and labeled them post hoc before running them. Holding each county's population denominator fixed at its 2013 value produces −1.684 (95 percent CI −5.035 to 1.666). Adding county-specific linear trends gives −5.370 (95 percent CI −12.916 to 2.175). Using startups per 10,000 residents as the outcome gives −0.530 (95 percent CI −1.177 to 0.118). Adding lagged population growth gives −1.386 (95 percent CI −4.715 to 1.942; $N = 895$). All of these intervals include zero; the diagnostics remain exploratory.

Across counties, the 2013-2022 change in branch density correlates 0.108 with log population growth and 0.224 with the change in branch count. Both the numerator and denominator matter, but these correlations do not explain why either changed.

6.2 Audit-added checks

The five-round audit added four checks after the original results were known, so none is confirmatory. The planned but omitted firm-age-zero firm rate gives −1.821 (95 percent CI −5.645 to 2.004; $N = 995$). Excluding the final BDS vintage gives −1.671 (95 percent CI −5.802 to 2.460; $N = 896$). Excluding Camden and Tyrrell, the two counties with any suppressed primary-outcome cells, gives −0.291 (95 percent CI −2.770 to 2.188; $N = 980$).

The timing check changes the interpretation most directly. It moves the SOD snapshot and controls from labels $t-1$ to $t-2$, placing the June snapshot before the March $t-1$ start of the BDS transition. The coefficient is **2.028** (county-clustered standard error 1.719; 95 percent CI −1.383 to 5.439; $p = 0.241$; $N = 895$). The estimate changes from negative to positive, while both confidence intervals remain wide and include zero. A4 also drops the first outcome year, so this comparison changes both timing and sample; it does not isolate timing alone. Neither specification supports a clear directional conclusion. Code-generated values appear in `output/tables/model_results_audit_additions.csv`.

7 Policy Interpretation and Limitations

North Carolina's traditional bank-office network contracted substantially. That descriptive result is clear. The relationship with employer-startup formation is less clear. The confidence intervals are broad, and moving branch measurements before the startup interval changes the sign of the point estimate. This uncertainty should not be read as evidence that branch closures are harmless.

The first limitation is measurement. Branch density is not access. Recent national geospatial work shows that large declines in branch counts can produce modest average changes in distance when

banks consolidate geographically redundant offices (Adams and Sherlund 2025). A county average can look stable even when a particular town loses its only convenient location. A stronger state monitoring system would combine exact branch locations with road-network travel time, population and employment grids, broadband, public transportation, credit-union offices, and options across county lines. This is a recommendation about measurement, not a claim that adding branches would raise startup rates.

The outcome also limits which businesses the analysis covers. Many entrepreneurs begin without payroll employees, but BDS excludes nonemployer businesses. Firm age zero is a useful public measure of new employer activity, not a measure of all business formation. Future work could compare BDS outcomes with applications from the Census Business Formation Statistics while preserving the distinction between an application and an operating employer firm.

Reverse causality remains the central identification problem. Banks choose locations using expectations about deposits, costs, population, and business demand. If banks close branches in counties where the economy is weakening, the fixed-effects slope mixes any effect of closure with the bank's response to local conditions. Closures can also occur in growing markets: a bank may close redundant offices in a strong urban market because digital use and nearby substitutes are greatest there. Either process can create a misleading county association. A stronger design would link exact closure events to merger-related consolidation exposure, test pretrends, and measure outcomes at a smaller geographic scale, closer to Nguyen (2019) and Ranish, Stella, and Zhang (2024).

Timing is also a material limitation. BDS flows labeled t cover the transition from March $t-1$ to March t, whereas the frozen SOD exposure is the June 30 snapshot labeled $t-1$. The headline exposure falls inside the outcome interval rather than cleanly before it. Moving the exposure to $t-2$ restores preperiod ordering but drops the first outcome year and leaves the design observational. The resulting positive, imprecise coefficient reverses the headline sign. Using an earlier snapshot improves temporal ordering but does not make branch placement exogenous.

The banking measures leave out important parts of the market. SOD excludes credit unions and nonbank lenders. Deposits are assigned to offices rather than customer locations, making county deposit HHI an imperfect measure of local borrower choice or credit supply. Online lenders and bank information technology may offset some changes in the physical network (Ahnert et al. 2024). I do not estimate historical community-bank share because the frozen pipeline lacks a defensible, time-consistent classification.

Precision and functional form matter as well. The panel has 100 county clusters and only ten outcome years. Confidence intervals are wide, especially for the metro and nonmetro comparison, and an interval containing zero does not prove the absence of a relationship. The Poisson result shows sensitivity to model form; the branch-count result shows sensitivity to how exposure is measured.

The practical implication concerns measurement and monitoring. North Carolina economic-development and banking organizations should track where branch losses alter practical access for small businesses rather than infer access from a statewide count or a per-capita ratio alone. Local evidence may reveal losses of access that a county average conceals. Any intervention should be

evaluated with outcomes and a design capable of separating policy effects from preexisting local trends.

8 Conclusion

This paper assembles a public panel for all 100 North Carolina counties over ten outcome years. The data verify a 21.8 percent decline in qualifying full-service branch offices from 2013 to 2022 and distinguish establishments created by new employer firms from all establishment entrants. The accompanying data, code, tables, and figures can be reproduced from federal sources.

The internally frozen model does not support the predicted positive association. Its branch-density coefficient is −1.823 startup establishments per 1,000 lagged establishments, with a 95 percent confidence interval from −5.412 to 1.767. Because the snapshot overlaps part of the BDS transition, this is not a clean lagged-exposure estimate. The audit-added clean-preperiod coefficient is 2.028, with a 95 percent confidence interval from −1.383 to 5.439. Rural interactions, alternative outcomes, and most robustness estimates are similarly imprecise. I retain the negative Poisson estimate as a model-dependent secondary result. Both timing models remain observational, and the estimates leave the direction and magnitude of the relationship unresolved.

The panel shows where more detailed evidence is needed. Branch counts contracted, but density is not distance, office presence is not credit supply, and fixed effects do not solve endogenous branch placement. A stronger next step would use geocoded, sub-county data, exact closure events, and plausibly exogenous merger-related variation. For now, the evidence supports careful monitoring of local access. It does not show that branch loss caused fewer startups, and it does not show that branch loss had no economic consequence.

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A Sample Construction and Disclosure Treatment

A.1 Sample flow

Stage	County-years
Raw BDS county rows, all states and years	146,924
NC rows, 2013-2023, including residual county code `999`	1,111
True NC county denominator rows after removing `999`	1,100
Candidate outcome panel, 2014-2023	1,000
Firm-age-zero rows in candidate window	1,000
Primary rows lost to age-zero disclosure suppression	5
Core banking/control/RUCC merge losses	0
Final primary estimation sample	**995**

Table A1. Sample flow. Source: code-generated `output/tables/table_sample_flow.csv` from Census BDS and merge audits. Negative loss counts in the code-generated table denote subtraction; this table presents those losses as positive counts.

A.2 Primary suppressed cells

FIPS	County	Outcome year	`age0_estabs` flag	Treatment
37029	Camden	2019	`D`	missing
37177	Tyrrell	2015	`D`	missing
37177	Tyrrell	2017	`D`	missing
37177	Tyrrell	2021	`D`	missing
37177	Tyrrell	2023	`D`	missing

Table A2. Disclosure-suppressed primary cells. `D` means disclosure suppression because too few firms are represented. No suppressed cell is converted to zero.

The cleaner also recognizes `S` for data-quality suppression, `X` for structurally zero or missing cells, and `N` for rates unavailable because their denominator is zero. Each flag is preserved in a separate field before numeric conversion. `data/interim/bds_flagged_cells.csv` records all flagged values encountered in retained fields.

B Variable Definitions and Summary Statistics

B.1 Core variables

Variable	Construction	Timing and units	Source
Employer-startup rate	1,000 × firm-age-zero establishments / lagged active establishments	outcome year; per 1,000	Census BDS
Official entry rate	published BDS establishment-entry rate	outcome year; BDS rate	Census BDS
Startup job intensity	1,000 × age-zero jobs created by births / lagged employment	outcome year; per 1,000 workers	Census BDS
Full-service branch density	10,000 × count of SOD codes 11/12 / population	June 30 labeled `t-1`; per 10,000 residents; overlaps the BDS March-to-March transition	FDIC SOD + BEA
Deposit HHI	10,000 × squared institution shares of branch-assigned deposits	June 30 labeled `t-1`; 0-10,000	FDIC SOD
Unemployment	annual county unemployment rate	label-aligned `t-1`; percent	BLS LAUS
Population / PCPI	county population and nominal per-capita income	label-aligned `t-1`; logs in model	BEA CAINC1
Nonmetro	2013 RUCC 4-9; 2023 vintage in robustness	time invariant	USDA ERS

Table B1. Core variable definitions. Full code-generated version: `output/tables/table_variable_definitions.csv`.

B.2 Summary statistics

Variable	N	Mean	SD	Median	Min	Max
Employer-startup rate	995	57.672	18.116	56.738	0.000	173.554
Age-zero establishments	995	139.806	317.176	52.000	0.000	2,761.000
Official BDS entry rate	997	8.983	2.191	8.848	2.670	18.533
Startup job intensity	995	20.271	13.177	18.036	0.000	154.175
Lagged branch density	1,000	2.449	1.071	2.322	0.489	8.373
Lagged branch count	1,000	22.904	35.788	12.000	1.000	249.000
Lagged deposit HHI	1,000	3,056.193	2,065.052	2,253.822	1,170.023	10,000.000
Lagged unemployment rate	1,000	5.772	1.886	5.400	3.100	15.000
Lagged population	1,000	102,253	164,751	52,793	3,234	1,174,107
Lagged PCPI	1,000	41,148	8,932	39,435	25,521	82,059

Table B2. Summary statistics. Source: code-generated `output/tables/table_summary_statistics.csv`. Statistics are unweighted county-year distributions.

C Complete Model Sequence

C.1 Main and heterogeneity estimates

ID	Outcome / effect	Estimate	SE	95% CI	p	N
M1	Startup rate, FE only	−1.418	1.866	[−5.121, 2.285]	0.449	995
M2	Startup rate, FE + controls	−1.823	1.809	[−5.412, 1.767]	0.316	995
M3a	Metro slope	1.575	2.921	[−4.220, 7.370]	0.591	995
M3b	Nonmetro minus metro	−3.888	2.846	[−9.534, 1.759]	0.175	995
M3c	Nonmetro slope	−2.312	1.818	[−5.919, 1.295]	0.206	995
M4	Official entry rate	−0.210	0.222	[−0.650, 0.230]	0.345	997
M5	Startup job intensity	−1.223	1.387	[−3.976, 1.529]	0.380	995

Table C1. Main estimates. SEs and confidence intervals are county-clustered. The interaction slopes are linear combinations from one model. Source: `output/tables/model_results_main.csv`.

C.2 Robustness estimates

ID	Specification	Estimate	SE	95% CI	p	N
R1	Exclude 2020-2021	−1.302	1.491	[−4.260, 1.656]	0.385	796
R2	Population weighted	−1.048	1.210	[−3.449, 1.352]	0.388	995
R3	Branch-count exposure	−0.001	0.157	[−0.313, 0.310]	0.994	995
R4	One-year count change	−0.130	0.217	[−0.559, 0.300]	0.551	995
R5	One-year density change	−4.216	2.385	[−8.948, 0.517]	0.080	995
R6	Deposit HHI / 1,000	0.636	1.078	[−1.503, 2.775]	0.556	995
R7	Establishment-exit context	0.280	0.233	[−0.183, 0.743]	0.233	998
R8a	2023-RUCC metro slope	2.368	1.949	[−1.499, 6.236]	0.227	995
R8b	2023-RUCC nonmetro slope	−2.462	1.897	[−6.226, 1.302]	0.197	995
R9	First differences	−3.749	2.703	[−9.113, 1.615]	0.169	891
R10	Poisson log slope	−0.040	0.018	[−0.077, −0.004]	0.028	995

Table C2. Predefined robustness estimates. Units differ across specifications; consult `output/tables/table_robustness_estimates.csv` for outcome and exposure labels. R10's incidence-rate ratio is 0.960. It is a secondary functional-form check, not the headline result.

C.3 Large-county influence

Leaving out Wake, Mecklenburg, Guilford, Forsyth, or Cumberland separately produces M2 slopes of −1.790, −1.790, −1.809, −1.811, and −1.821. Every confidence interval includes zero. No single one of these five counties accounts for the headline point estimate.

D Post-hoc Diagnostic Boundary

The post-hoc models were added only after the frozen branch-density estimate and branch-count estimate had been inspected. Their purpose is diagnosis, not confirmation.

ID	Diagnostic	Estimate	SE	95% CI	p	N
P1	Density with 2013 population fixed	−1.684	1.689	[−5.035, 1.666]	0.321	995
P2	County-specific linear trends	−5.370	3.803	[−12.916, 2.175]	0.161	995
P3	Startups per 10,000 residents outcome	−0.530	0.326	[−1.177, 0.118]	0.108	995
P4	Additional population-growth control	−1.386	1.677	[−4.715, 1.942]	0.411	895

Table D1. Exploratory diagnostics. Source: `output/tables/model_results_posthoc_diagnostics.csv`. These estimates were added after I inspected the original results and are outside the internally frozen sequence.

D.1 Five-round audit additions

The following checks were estimated during the final audit, after the original results were known. A1 fills an omission in the original run: the internally frozen plan listed the firm-age-zero firm rate, but the original model sequence did not estimate it. A2 and A3 test vintage and suppression sensitivity. A4 corrects the exposure timing by moving the SOD snapshot and controls back one additional label year so that June `t-2` precedes the March `t-1` start of the BDS outcome transition.

ID	Audit-added check	Estimate	SE	95% CI	p	N
A1	Firm-age-zero firms per 1,000 lagged firms	−1.821	1.928	[−5.645, 2.004]	0.347	995
A2	Exclude final BDS outcome year	−1.671	2.082	[−5.802, 2.460]	0.424	896
A3	Exclude Camden and Tyrrell	−0.291	1.249	[−2.770, 2.188]	0.816	980
A4	Clean-preperiod exposure and controls (`t-2`)	2.028	1.719	[−1.383, 5.439]	0.241	895

Table D2. Audit-added checks. Source: `output/tables/model_results_audit_additions.csv`. These estimates are diagnostic, not confirmatory. Moving the exposure back another year in A4 changes the sign, but the wide interval does not establish a positive relationship.

E Data and Model Quality Assurance

The automated integrity script checks:

- 1,000 panel rows and 100 counties;
- unique (`fips`, `year`) keys;
- five-character NC FIPS values;
- `exposure_year = year - 1` for every row;
- 995 primary observed outcomes;
- 100 true BDS counties in every retained year;
- positive lagged denominators;
- flagged age-zero cells remain missing;
- only recognized BDS flags;
- nonnegative branch counts and plausible density range;
- positive population and PCPI;
- unemployment within 0-100;
- 1,100 FDIC county-year skeleton rows;
- zero missing core merged variables; and
- a SHA-256 match for every raw source file.

All checks pass in `docs/quality_control_report.md`.

The original independent verification bypassed the formula/regression helper for M2. It constructed an explicit 995×113 design matrix containing an intercept, four continuous regressors, 99 county dummies, and nine year dummies. NumPy least squares reproduced the M2 branch-density coefficient to within 4.1×10^{-14}. A manually constructed CR1 county-clustered sandwich covariance reproduced the stored standard error within 7.3×10^{-12}. The matrix had full rank 113 and 100 clusters.

The five-round audit extended that check to all 31 reported main, interaction, robustness, influence, post-hoc, and audit-added effects. It rebuilt each fixed-effects matrix directly, constructed CR1 cluster covariance independently, and fitted the Poisson model with a separate iteratively reweighted least-squares routine. All 31 effects passed a 10^{-8} tolerance. The maximum absolute differences were 3.73×10^{-14} for coefficients and 2.11×10^{-9} for standard errors. Full details are in `docs/independent_full_model_audit.md` and `output/tables/full_result_independent_audit.csv`.

F Reproducibility and Claim Boundaries

The acquisition script records exact government URLs, file names, access date, release coverage, intended use, limitations, and checksums. Cleaning, merging, analysis, table generation, and figure generation are separate scripts. Generated datasets, tables, and figures are not manually altered.

The following statements are supported by this package:

1. The qualifying NC full-service branch count fell from 2,554 in 2013 to 1,998 in 2022.
2. The constructed panel contains 1,000 candidate county-years and 995 primary observations.
3. The headline fixed-effects estimate, uncertainty, and robustness values reported in the paper match code-generated tables.
4. Firm-age-zero establishments distinguish new employer-firm entry from all establishment entrants under the Census definition.
5. The frozen label-aligned exposure partly overlaps the March-to-March BDS outcome interval.
6. The clean-preperiod estimate is positive and imprecise, so the sign is not temporally stable.

The following statements are **not** supported:

1. Branch closures caused fewer or more startups.
2. Branch density measures actual geographic or financial access.
3. The results apply to nonemployer businesses.
4. Deposit HHI measures the full county credit market.
5. Metro and nonmetro effects differ.
6. A conventional p-value in the Poisson model establishes a real causal mechanism.
7. The manuscript is peer reviewed, published, accepted, or under review.